

■ CLASS V ENGLISH QUESTION BANK

CHAPTER 05 QUESTION BANK (300 QUESTIONS)

Subject: English	Class: V
Chapter: 05	Total Questions: 300 (50 per category across 6 files)
Source: Comprehension Book 5	Board / Blueprint: ICSE / CBSE Class V English Pattern

1. Multiple Choice Questions (MCQs)

> **Category:** Multiple Choice Questions | **Total:** 50 Questions | **Marks:** 1 each

Question 1

Question ID: BK05_CH05_MCQ_001

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: History Concept

Marks: 1

Question: Was the computer invented by a single person?

- (A) No, it was developed over many years by several brilliant minds
- (B) Yes, Charles Babbage built it all alone
- (C) Yes, Steve Jobs invented it in one day
- (D) Yes, Alan Turing built it in his garage

Answer Key: **(A)** — Developed over many years by several brilliant minds.

Question 2

Question ID: BK05_CH05_MCQ_002

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Pioneers

Marks: 1

Question: Who designed the Difference Engine in 1822?

- (A) Charles Babbage
- (B) Ada Lovelace
- (C) Alan Turing

- (D) John Mauchly

Answer Key: **(A) — Charles Babbage designed the Difference Engine in 1822.**

Question 3

Question ID: BK05_CH05_MCQ_003

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Machine Function

Marks: 1

Question: What primary task was the Difference Engine designed to perform?

- (A) Mathematical calculations
- (B) Send text messages
- (C) Play audio music
- (D) Display color photographs

Answer Key: **(A) — Designed to perform mathematical calculations.**

Question 4

Question ID: BK05_CH05_MCQ_004

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Concepts

Marks: 1

Question: What was the name of the advanced machine planned by Charles Babbage that is considered the first concept of a modern computer?

- (A) The Analytical Engine
- (B) The Difference Engine
- (C) The ENIAC
- (D) The Apple I

Answer Key: **(A) — The Analytical Engine is considered the first concept of a modern computer.**

Question 5

Question ID: BK05_CH05_MCQ_005

Type: MCQ

Difficulty: Easy

Bloom Level: Understanding

Topic: Historical Barrier

Marks: 1

Question: Why could Charles Babbage not complete his Analytical Engine?

- (A) Technology at the time was not advanced enough
- (B) He ran out of paper
- (C) He lost his interest in mathematics

- (D) The King of England forbade it

Answer Key: **(A) — Technology at the time was not advanced enough.**

Question 6

Question ID: BK05_CH05_MCQ_006

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Pioneers

Marks: 1

Question: Who is known as the world's first computer programmer?

- (A) Ada Lovelace
- (B) Charles Babbage
- (C) Alan Turing
- (D) John Presper Eckert

Answer Key: **(A) — Ada Lovelace wrote the first instructions/algorithms.**

Question 7

Question ID: BK05_CH05_MCQ_007

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Achievement

Marks: 1

Question: What did Ada Lovelace write for Babbage's Analytical Engine?

- (A) The first computer instructions or algorithms
- (B) The first science fiction novel
- (C) A dictionary of numbers
- (D) A piano music composition

Answer Key: **(A) — She wrote the first instructions or algorithms.**

Question 8

Question ID: BK05_CH05_MCQ_008

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: World War II

Marks: 1

Question: What machine did British scientist Alan Turing create during World War II?

- (A) A machine to break secret codes
- (B) The first digital pocket watch
- (C) A steam-powered calculator

- (D) A sound recording machine

Answer Key: **(A) — Alan Turing created a machine that could break secret codes.**

Question 9

Question ID: BK05_CH05_MCQ_009

Type: MCQ

Difficulty: Easy

Bloom Level: Understanding

Topic: Theoretical Impact

Marks: 1

Question: What major theoretical field did Alan Turing's ideas help lay the foundation for?

- (A) Modern computers and thinking machines
- (B) Steam engine mechanics
- (C) Radio tower engineering
- (D) Chemical medicine

Answer Key: **(A) — His ideas laid foundations for modern computers.**

Question 10

Question ID: BK05_CH05_MCQ_010

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Milestones

Marks: 1

Question: What was the name of the first electronic computer built in 1945?

- (A) ENIAC
- (B) Microchip
- (C) Analytical Engine
- (D) Difference Engine

Answer Key: **(A) — ENIAC (Electronic Numerical Integrator and Computer) built in 1945.**

Question 11

Question ID: BK05_CH05_MCQ_011

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Inventors

Marks: 1

Question: Who built the first electronic computer ENIAC in 1945?

- (A) John Presper Eckert and John Mauchly
- (B) Charles Babbage and Ada Lovelace
- (C) Alan Turing and Bill Gates

- (D) Steve Jobs and Steve Wozniak

Answer Key: **(A) — Built by John Presper Eckert and John Mauchly in the United States.**

Question 12

Question ID: BK05_CH05_MCQ_012

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Physical Size

Marks: 1

Question: How large was the ENIAC computer when it was built?

- (A) Mammoth, filling an entire room
- (B) Small enough to fit on a wrist
- (C) The size of a handheld book
- (D) The size of a postage stamp

Answer Key: **(A) — ENIAC was mammoth, filling an entire room.**

Question 13

Question ID: BK05_CH05_MCQ_013

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Performance

Marks: 1

Question: How did ENIAC's calculation speed compare to human speed?

- (A) It could perform calculations much faster than any human
- (B) It was ten times slower than a human
- (C) It made errors every minute
- (D) It calculated at the exact same speed as a child

Answer Key: **(A) — It performed calculations much faster than any human.**

Question 14

Question ID: BK05_CH05_MCQ_014

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: 1970s Breakthrough

Marks: 1

Question: Which 1970s invention made personal computers possible?

- (A) The microchip
- (B) The mechanical disc
- (C) The vacuum tube

- (D) The telegraph key

Answer Key: **(A) — The invention of the microchip in the 1970s made personal computers possible.**

Question 15

Question ID: BK05_CH05_MCQ_015

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Tech Companies

Marks: 1

Question: Which companies helped bring computers into homes, schools, and offices around the world?

- (A) Apple, IBM, and Microsoft
- (B) Ford, Boeing, and General Motors
- (C) Sony, Nintendo, and Sega
- (D) Tesla, Space-X, and NASA

Answer Key: **(A) — Companies like Apple, IBM, and Microsoft helped bring computers into homes.**

Question 16

Question ID: BK05_CH05_MCQ_016

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Ubiquity

Marks: 1

Question: Where can computers be found today according to Chapter 05?

- (A) On our desks, in our pockets, and even in watches
- (B) Only in secret underground bunkers
- (C) Only on large ocean ships
- (D) Only inside university laboratories

Answer Key: **(A) — Computers are on desks, in pockets, and in watches.**

Question 17

Question ID: BK05_CH05_MCQ_017

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Acronym

Marks: 1

Question: What does the acronym ENIAC stand for?

- (A) Electronic Numerical Integrator and Computer
- (B) Electrical New Invention And Code
- (C) Engine Numbers In Automatic Calculation

- (D) Electronic National Information And Circuit

Answer Key: **(A) — ENIAC stands for Electronic Numerical Integrator and Computer.**

Question 18

Question ID: BK05_CH05_MCQ_018

Type: MCQ

Difficulty: Easy

Bloom Level: Understanding

Topic: Vocabulary

Marks: 1

Question: What does the word 'mammoth' mean in the vocabulary box?

- (A) Very big or huge in size
- (B) Very tiny or microscopic
- (C) Made of solid wood
- (D) Moving extremely fast

Answer Key: **(A) — Mammoth means very big.**

Question 19

Question ID: BK05_CH05_MCQ_019

Type: MCQ

Difficulty: Easy

Bloom Level: Understanding

Topic: Vocabulary

Marks: 1

Question: What does the word 'integrator' mean?

- (A) Something which joins things so that they become one
- (B) A tool used to cut metal wires
- (C) A device that makes sounds quiet
- (D) A mathematical zero

Answer Key: **(A) — Integrator means something joining things so they become one.**

Question 20

Question ID: BK05_CH05_MCQ_020

Type: MCQ

Difficulty: Easy

Bloom Level: Understanding

Topic: Vocabulary

Marks: 1

Question: What does the word 'algorithm' mean?

- (A) A step-by-step set of instructions for solving a problem
- (B) A physical gear inside a clock
- (C) A secret language used by birds

- (D) A type of computer monitor screen

Answer Key: **(A) — Algorithm means step-by-step instructions for solving a problem.**

Question 21

Question ID: BK05_CH05_MCQ_021

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Century

Marks: 1

Question: In which century did Charles Babbage begin designing computing engines?

- (A) Early 19th century
- (B) 15th century
- (C) 21st century
- (D) 12th century

Answer Key: **(A) — The journey began in the early 19th century.**

Question 22

Question ID: BK05_CH05_MCQ_022

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Dates

Marks: 1

Question: In which year was Charles Babbage's Difference Engine designed?

- (A) 1822
- (B) 1750
- (C) 1945
- (D) 1999

Answer Key: **(A) — Difference Engine was designed in 1822.**

Question 23

Question ID: BK05_CH05_MCQ_023

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Historical Event

Marks: 1

Question: What conflict led Alan Turing to build code-breaking machines?

- (A) World War II
- (B) World War I
- (C) The American Revolution

- (D) The Napoleonic Wars

Answer Key: *(A)*** — Alan Turing created code-breaking machines during World War II.**

Question 24

Question ID: BK05_CH05_MCQ_024

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Evolution Trend

Marks: 1

****Question**:** How have computers changed over time regarding size and cost?

- (A) They became smaller, faster, and more affordable
- (B) They became bigger, slower, and more expensive
- (C) They remained identical in size and price
- (D) They disappeared from human society

Answer Key: *(A)*** — Over time, computers became smaller, faster, and more affordable.**

Question 25

Question ID: BK05_CH05_MCQ_025

Type: MCQ

Difficulty: Easy

Bloom Level: Remembering

Topic: Chapter Title

Marks: 1

****Question**:** What title is given to Chapter 05?

- (A) The Invention of the Computer
- (B) The Invention of Television
- (C) The Missile Man of India
- (D) The Narmada River

Answer Key: *(A)*** — Title is 'The Invention of the Computer'.**

Question 26

Question ID: BK05_CH05_MCQ_026

Type: MCQ

Difficulty: Medium

Bloom Level: Analyzing

Topic: Historical Significance

Marks: 1

****Question**:** Why is Charles Babbage called the 'Father of Modern Computing' despite not finishing the Analytical Engine?

- (A) His Analytical Engine design established the fundamental logical architecture used in modern computers
- (B) He built the first internet network
- (C) He founded Apple and Microsoft

- (D) He invented electricity

Answer Key: **(A) — His Analytical Engine established fundamental logical computer architecture.**

Question 27

Question ID: BK05_CH05_MCQ_027

Type: MCQ

Difficulty: Medium

Bloom Level: Evaluating

Topic: Lovelace Contribution

Marks: 1

Question: What was the unique significance of Ada Lovelace's contribution to computing?

- (A) She realized machines could process symbols and data beyond pure arithmetic by using programmed algorithms
- (B) She manufactured the metal gears for Babbage
- (C) She invented the computer mouse
- (D) She wrote laws governing computer security

Answer Key: **(A) — Realized machines could process symbolic data via algorithms.**

Question 28

Question ID: BK05_CH05_MCQ_028

Type: MCQ

Difficulty: Medium

Bloom Level: Analyzing

Topic: Military Impact

Marks: 1

Question: How did World War II accelerate computer science through Alan Turing's work?

- (A) Urgent military necessity required automated code-breaking, driving creation of early programmable machines
- (B) Soldiers used computers to send emails
- (C) Computers were used to fly airplanes automatically
- (D) WWII stopped all scientific research

Answer Key: **(A) — Urgent military code-breaking required automated programmable machines.**

Question 29

Question ID: BK05_CH05_MCQ_029

Type: MCQ

Difficulty: Medium

Bloom Level: Comparing

Topic: Technical Evolution

Marks: 1

Question: What made ENIAC a true 'electronic' computer unlike Babbage's engines?

- (A) ENIAC used vacuum tubes and electrical pulses instead of mechanical gears and rods to perform calculations

- (B) ENIAC ran on steam power
- (C) ENIAC used solar energy
- (D) Babbage used microchips while ENIAC used wood

Answer Key: **(A) — Used vacuum tubes and electrical pulses instead of mechanical gears.**

Question 30

Question ID: BK05_CH05_MCQ_030

Type: MCQ

Difficulty: Medium

Bloom Level: Analyzing

Topic: Microchip Revolution

Marks: 1

Question: How did the microchip transform the computer industry in the 1970s?

- (A) By shrinking millions of electronic circuits onto tiny silicon chips, allowing computers to become portable and affordable
- (B) By making computers run on water
- (C) By replacing computer screens with paper
- (D) By making computers room-sized again

Answer Key: **(A) — Shrank circuits onto tiny chips, enabling portable, affordable personal computers.**

Question 31

Question ID: BK05_CH05_MCQ_031

Type: MCQ

Difficulty: Medium

Bloom Level: Understanding

Topic: Commercial Impact

Marks: 1

Question: What role did companies like Apple, IBM, and Microsoft play in computer history?

- (A) They commercialized personal computer hardware and user-friendly software for homes, schools, and businesses
- (B) They invented mechanical engines in 1822
- (C) They built code-breaking machines for World War II
- (D) They manufactured radio towers

Answer Key: **(A) — Commercialized PC hardware and software for homes, schools, and offices.**

Question 32

Question ID: BK05_CH05_MCQ_032

Type: MCQ

Difficulty: Medium

Bloom Level: Comparing

Topic: Footprint Comparison

Marks: 1

Question: Compare the physical footprint of ENIAC (1945) with a modern smartphone.

- (A) ENIAC filled an entire 1,800 sq ft room, whereas a modern smartphone fits in a pocket yet is exponentially faster
- (B) ENIAC was smaller than a smartphone
- (C) Both machines weighed 30 tons
- (D) Smartphones fill an entire classroom

Answer Key: **(A) — ENIAC filled a room (30 tons), whereas a smartphone fits in a pocket.**

Question 33

Question ID: BK05_CH05_MCQ_033

Type: MCQ

Difficulty: Medium

Bloom Level: Understanding

Topic: Software Concept

Marks: 1

Question: Why is computer programming considered essential for computer operation?

- (A) Hardware cannot perform tasks without software algorithms telling it step-by-step what calculations to execute
- (B) Computers work automatically without any instructions
- (C) Programming is only used to color the computer case
- (D) Hardware creates its own algorithms automatically

Answer Key: **(A) — Hardware requires software algorithms to direct calculation steps.**

Question 34

Question ID: BK05_CH05_MCQ_034

Type: MCQ

Difficulty: Medium

Bloom Level: Analyzing

Topic: Hardware Context

Marks: 1

Question: What does the word 'mammoth' illustrate about 1940s computer hardware?

- (A) Thousands of large vacuum tubes, switches, and heavy cables required massive physical space
- (B) Computers were made from mammoth animal bones
- (C) 1940s computers were weak and tiny
- (D) Vacuum tubes were invisible to human eyes

Answer Key: **(A) — Thousands of vacuum tubes and cables required massive room space.**

Question 35

Question ID: BK05_CH05_MCQ_035

Type: MCQ

Difficulty: Medium

Bloom Level: Evaluating

Topic: Societal Impact

Marks: 1

****Question**:** How do computers impact daily human productivity today?

- (A) By automating complex calculations, storing vast data, enabling instant communication, and providing digital learning
- (B) By forcing people to do all math manually
- (C) By eliminating the need for electricity
- (D) By replacing all human thoughts

Answer Key: ** (A) **** — Automating calculations, storing data, enabling communication, and digital learning.**

Question 36

Question ID: BK05_CH05_MCQ_036

Type: MCQ

Difficulty: Medium

Bloom Level: Understanding

Topic: Engineering Constraint

Marks: 1

****Question**:** What limitation prevented 19th-century engineers from building Babbage's Analytical Engine?

- (A) Precision machining tools were not precise enough to manufacture thousands of intricate metal gears consistently
- (B) Paper was not invented yet
- (C) Engineers did not know mathematics
- (D) Babbage hid his design blueprints

Answer Key: ** (A) **** — Precision machining could not manufacture thousands of delicate gears consistently.**

Question 37

Question ID: BK05_CH05_MCQ_037

Type: MCQ

Difficulty: Medium

Bloom Level: Evaluating

Topic: Turing & AI

Marks: 1

****Question**:** Why is Alan Turing considered a founding father of Artificial Intelligence (AI)?

- (A) He conceptualized 'thinking machines' and proposed criteria for testing machine intelligence (Turing Test)
- (B) He built robots that could walk and talk
- (C) He created the first internet website
- (D) He invented computer video games

Answer Key: ** (A) **** — Conceptualized thinking machines and criteria for machine intelligence.**

Question 38

Question ID: BK05_CH05_MCQ_038

Type: MCQ

Difficulty: Medium

Bloom Level: Comparing

Topic: Device Comparison

Marks: 1

****Question**:** What is the primary difference between a calculator and a general-purpose computer?

- (A) A calculator performs specific fixed math; a general-purpose computer can run unlimited different programmed algorithms
- (B) A calculator uses electricity while a computer does not
- (C) A calculator is bigger than ENIAC
- (D) A computer can only add numbers 1 to 10

Answer Key: **A)**** — Computer runs unlimited programmed algorithms; calculator performs fixed math.**

Question 39

Question ID: BK05_CH05_MCQ_039

Type: MCQ

Difficulty: Medium

Bloom Level: Analyzing

Topic: Postwar Transfer

Marks: 1

****Question**:** How did World War II code-breaking machines influence postwar civilian computing?

- (A) Wartime breakthroughs in high-speed electronic circuits directly informed the design of commercial computers like ENIAC
- (B) Code-breaking machines were destroyed and forgotten
- (C) Wartime machines were converted into television sets
- (D) Postwar computing returned to mechanical clockwork

Answer Key: **A)**** — Wartime electronic circuit breakthroughs informed commercial computer design.**

Question 40

Question ID: BK05_CH05_MCQ_040

Type: MCQ

Difficulty: Medium

Bloom Level: Evaluating

Topic: Pedagogical Insight

Marks: 1

****Question**:** What lesson about collaborative innovation does Chapter 05 teach Class 5 students?

- (A) Major technological revolutions emerge from generations of inventors building upon each other's ideas over centuries
- (B) One genius invents everything without help
- (C) Innovation stops once a machine is built
- (D) Collaboration slows down scientific discovery

Answer Key: **A)**** — Revolutions emerge from generations of inventors building upon each other.**

Question 41

Question ID: BK05_CH05_MCQ_041

Type: MCQ

Difficulty: Hard

Bloom Level: Evaluating

Topic: HOTS Computing Architecture

Marks: 1

****Question**:** Critique the transition from mechanical decimal computing (Babbage) to electronic binary computing (ENIAC/Turing).

- (A) Decimal mechanical gears were physically complex and slow, whereas binary electronic circuits (on/off voltage) allowed ultra-fast, reliable calculation
- (B) Decimal gears were faster than electronic voltage
- (C) Binary computing requires mechanical water wheels
- (D) Vacuum tubes only worked with decimal numbers

Answer Key: ** (A) **** — Binary electronic circuits (on/off voltage) allowed ultra-fast, reliable calculations over physical gears.**

Question 42

Question ID: BK05_CH05_MCQ_042

Type: MCQ

Difficulty: Hard

Bloom Level: Analyzing

Topic: Moore's Law Analysis

Marks: 1

****Question**:** Deconstruct Moore's Law as presaged by the 1970s microchip revolution in Chapter 05.

- (A) Shrinking transistor size on microchips exponentially increased computer processing power while dramatically reducing cost and size
- (B) Microchips made computers ten times larger every decade
- (C) Transistors were replaced by mechanical clockwork in 1970
- (D) Computer power decreases as technology advances

Answer Key: ** (A) **** — Shrinking transistor size exponentially increased power while reducing cost and physical size.**

Question 43

Question ID: BK05_CH05_MCQ_043

Type: MCQ

Difficulty: Hard

Bloom Level: Evaluating

Topic: Ethical Evaluation

Marks: 1

****Question**:** Evaluate the ethical and societal responsibility stemming from ubiquitous computing today.

- (A) Universal computer access requires protecting data privacy, preventing digital divides, and promoting responsible technology use
- (B) Computers should be restricted to university professors only
- (C) Data privacy is unimportant in modern society
- (D) Computers eliminate all ethical responsibilities

Answer Key: ** (A) **** — Requires data privacy protection, bridging digital divides, and responsible usage.**

Question 44

Question ID: BK05_CH05_MCQ_044

Type: MCQ

Difficulty: Hard

Bloom Level: Comparing

Topic: Comparative Logic Analysis

Marks: 1

****Question**:** Compare Ada Lovelace's 1840s algorithmic concepts with modern software engineering.

- (A) Lovelace's loop and branching concepts for the Analytical Engine remain the foundational logic structures of modern coding languages
- (B) Lovelace's algorithms were written in Python
- (C) Modern software engineering uses no logic structures
- (D) Lovelace's work had no relation to programming logic

Answer Key: ** (A) **** — Lovelace's loop and branching logic concepts remain foundations of modern coding.**

Question 45

Question ID: BK05_CH05_MCQ_045

Type: MCQ

Difficulty: Hard

Bloom Level: Creating

Topic: Future Computing Vision

Marks: 1

****Question**:** Formulate a vision of future quantum or biological computing for Class 5 students.

- (A) 'Future computers may use quantum bits or DNA molecules to solve complex climate and medical problems in seconds.'
- (B) 'Future computers will return to room-sized vacuum tube boxes.'
- (C) 'Computers will be replaced by mechanical abacuses.'
- (D) 'Calculations will no longer be needed in science.'

Answer Key: ** (A) **** — Vision of quantum bits and bio-molecules solving complex global challenges.**

Question 46

Question ID: BK05_CH05_MCQ_046

Type: MCQ

Difficulty: Hard

Bloom Level: Evaluating

Topic: Historical Funding Assessment

Marks: 1

****Question**:** Assess the historical role of government funding in wartime computing breakthroughs.

- (A) Government funding during World War II provided resources for Turing and ENIAC teams, accelerating technology by decades
- (B) Wartime funding slowed down computer research
- (C) Governments banned computer development during wars

- (D) Computing breakthroughs occurred without any funding

Answer Key: **(A) — Wartime funding accelerated computing breakthroughs by decades.**

Question 47

Question ID: BK05_CH05_MCQ_047

Type: MCQ

Difficulty: Hard

Bloom Level: Analyzing

Topic: User Interface Analysis

Marks: 1

Question: Analyze how GUI (Graphical User Interface) developed by Apple/Microsoft transformed user accessibility.

- (A) GUIs replaced complex text command lines with visual icons and mouse clicks, making computers usable by non-experts
- (B) GUIs made computers harder to use for ordinary people
- (C) GUIs eliminated color screens
- (D) GUIs forced users to write binary code manually

Answer Key: **(A) — Icons and mouse clicks made computers accessible to non-technical users.**

Question 48

Question ID: BK05_CH05_MCQ_048

Type: MCQ

Difficulty: Hard

Bloom Level: Synthesizing

Topic: Cross-Disciplinary Synthesis

Marks: 1

Question: Synthesize how Chapter 05 integrates mathematics, history, and computer science concepts.

- (A) Combines mathematical origins (Babbage/Lovelace) with historical contexts (WWII/1970s) and computer science principles (algorithms/hardware)
- (B) Eliminates history to focus on arithmetic
- (C) Focuses solely on spelling computer terms
- (D) Replaces computer science with ancient geography

Answer Key: **(A) — Integrates math origins, historical context, and computer science principles.**

Question 49

Question ID: BK05_CH05_MCQ_049

Type: MCQ

Difficulty: Hard

Bloom Level: Evaluating

Topic: Historical Accuracy Critique

Marks: 1

Question: Critique the claim: 'ENIAC was the world's first personal computer.'

- (A) Inaccurate; ENIAC was a room-sized, room-filling multi-user military calculation machine; personal computers emerged only in the 1970s microchip era
- (B) Completely true; ENIAC was built for home desks
- (C) False; ENIAC was a small pocket calculator
- (D) True; ENIAC was sold in retail stores in 1945

Answer Key: **(A) — Inaccurate; ENIAC was a mammoth room-sized machine; PCs emerged in the 1970s microchip era.**

Question 50

Question ID: BK05_CH05_MCQ_050

Type: MCQ

Difficulty: Hard

Bloom Level: Creating

Topic: Assessment Task Design

Marks: 1

Question: Formulate a comprehensive essay prompt based on Chapter 05 for a Class 5 assessment.

- (A) 'Trace the development of the computer from Babbage's mechanical engines to modern microchips. Highlight the contributions of Babbage, Lovelace, Turing, and ENIAC.'
- (B) 'Write five sentences about your favorite video game.'
- (C) 'List ten brands of laptop computers.'
- (D) 'Draw a picture of a keyboard.'

Answer Key: **(A) — Comprehensive essay prompt evaluating historical chronology, technical milestones, and pioneer contributions.**

2. Fill in the Blanks

> **Category**: Fill in the Blanks | **Total**: 50 Questions | **Marks**: 1 each

Question 1

Question ID: BK05_CH05_FIB_001

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: The computer was not invented by one person but developed over many _____.

Answer Key: years (Developed over many years.)

Question 2

Question ID: BK05_CH05_FIB_002

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Charles Babbage designed the Difference Engine in the year _____.

Answer Key: 1822 (Designed in 1822.)

Question 3

Question ID: BK05_CH05_FIB_003

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Charles Babbage planned an advanced machine called the _____ Engine.

Answer Key: Analytical (Planned the Analytical Engine.)

Question 4

Question ID: BK05_CH05_FIB_004

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Babbage could not complete the Analytical Engine because _____ was not advanced enough.

Answer Key: technology (Technology was not advanced enough.)

Question 5

Question ID: BK05_CH05_FIB_005

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Ada Lovelace is recognized as the world's first computer _____.

Answer Key: programmer (World's first computer programmer.)

Question 6

Question ID: BK05_CH05_FIB_006

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Ada Lovelace wrote the first instructions or _____ for Babbage's machine.

Answer Key: algorithms (Wrote first instructions or algorithms.)

Question 7

Question ID: BK05_CH05_FIB_007

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Alan Turing created a machine during World War II to break secret _____.

Answer Key: codes (Machine to break secret codes.)

Question 8

Question ID: BK05_CH05_FIB_008

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: The first electronic computer was called _____.

Answer Key: ENIAC (First electronic computer was ENIAC.)

Question 9

Question ID: BK05_CH05_FIB_009

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: ENIAC was built in the year _____.

Answer Key: 1945 (Built in 1945.)

Question 10

Question ID: BK05_CH05_FIB_010

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: ENIAC was built by John Presper Eckert and John _____ in the United States.

Answer Key: Mauchly (Built by Eckert and Mauchly.)

Question 11

Question ID: BK05_CH05_FIB_011

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: ENIAC was _____, filling an entire room.

Answer Key: **mammoth** (ENIAC was mammoth.)

Question 12

Question ID: BK05_CH05_FIB_012

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: The invention of the _____ in the 1970s made personal computers possible.

Answer Key: **microchip** (Microchip in 1970s.)

Question 13

Question ID: BK05_CH05_FIB_013

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Companies like Apple, IBM, and _____ brought computers into homes.

Answer Key: **Microsoft** (Apple, IBM, and Microsoft.)

Question 14

Question ID: BK05_CH05_FIB_014

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Today, computers can be found on our desks, in our pockets, and even in _____.

Answer Key: **watches** (Found in watches.)

Question 15

Question ID: BK05_CH05_FIB_015

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: An integrator is defined as something which joins things so they become _____.

Answer Key: **one** (Joins things so they become one.)

Question 16

Question ID: BK05_CH05_FIB_016

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Mammoth means very _____.

Answer Key: **big** (Mammoth means very big.)

Question 17

Question ID: BK05_CH05_FIB_017

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Charles Babbage was a British _____.

Answer Key: mathematician (Babbage was a British mathematician.)

Question 18

Question ID: BK05_CH05_FIB_018

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Alan Turing was a British _____.

Answer Key: scientist (Alan Turing was a scientist.)

Question 19

Question ID: BK05_CH05_FIB_019

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: The Difference Engine was built to perform mathematical _____.

Answer Key: calculations (Perform mathematical calculations.)

Question 20

Question ID: BK05_CH05_FIB_020

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: ENIAC stands for Electronic Numerical Integrator and _____.

Answer Key: Computer (Integrator and Computer.)

Question 21

Question ID: BK05_CH05_FIB_021

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Personal computers entered homes, schools, and _____ around the world.

Answer Key: offices (Homes, schools, and offices.)

Question 22

Question ID: BK05_CH05_FIB_022

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Ada Lovelace worked closely with mathematician Charles _____.

Answer Key: Babbage (Worked with Charles Babbage.)

Question 23

Question ID: BK05_CH05_FIB_023

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Alan Turing's ideas laid the foundation for modern _____.

Answer Key: computers (Foundation for modern computers.)

Question 24

Question ID: BK05_CH05_FIB_024

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Microchips shrunk electronic circuits onto tiny silicon _____.

Answer Key: wafers (Shrunk circuits onto silicon wafers.)

Question 25

Question ID: BK05_CH05_FIB_025

Type: Fill in the Blanks

Difficulty: Easy

Marks: 1

Question: Chapter 05 is titled 'The Invention of the _____'.

Answer Key: Computer (Titled 'The Invention of the Computer'.)

Question 26

Question ID: BK05_CH05_FIB_026

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Babbage's Analytical Engine established the conceptual framework for computer _____.

Answer Key: architecture (Established framework for computer architecture.)

Question 27

Question ID: BK05_CH05_FIB_027

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Algorithms provide logical step-by-step guidance for software _____.

Answer Key: execution (Guidance for software execution.)

Question 28

Question ID: BK05_CH05_FIB_028

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Wartime code-breaking accelerated developments in electronic digital _____.

Answer Key: *circuitry* (Accelerated developments in electronic circuitry.)

Question 29

Question ID: BK05_CH05_FIB_029

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: ENIAC utilized thousands of vacuum tubes for high-speed numerical _____.

Answer Key: *computation* (Vacuum tubes for numerical computation.)

Question 30

Question ID: BK05_CH05_FIB_030

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Silicon microchips allowed millions of transistors to fit on one _____.

Answer Key: *chip* (Transistors fit on one chip.)

Question 31

Question ID: BK05_CH05_FIB_031

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Personal computing transformed daily education, office work, and global _____.

Answer Key: *communication* (Transformed work and communication.)

Question 32

Question ID: BK05_CH05_FIB_032

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Analytical Engine designs included an arithmetic unit and built-in _____.

Answer Key: *memory* (Included arithmetic unit and memory.)

Question 33

Question ID: BK05_CH05_FIB_033

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Lovelace envisioned computers processing non-numerical data such as _____.

Answer Key: *music* (Envisioned processing music and symbols.)

Question 34

Question ID: BK05_CH05_FIB_034

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Turing's theoretical model laid the groundwork for artificial _____.

Answer Key: **intelligence (Groundwork for artificial intelligence.)**

Question 35

Question ID: BK05_CH05_FIB_035

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Miniaturization reduced computer sizes from room-scale to hand-held _____.

Answer Key: **devices (Reduced size to hand-held devices.)**

Question 36

Question ID: BK05_CH05_FIB_036

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Early computers relied on punch cards for data input and program _____.

Answer Key: **loading (Punch cards for program loading.)**

Question 37

Question ID: BK05_CH05_FIB_037

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Postwar commercial computers automated business inventory and financial _____.

Answer Key: **accounting (Automated business accounting.)**

Question 38

Question ID: BK05_CH05_FIB_038

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: User-friendly graphical interfaces simplified complex computer _____.

Answer Key: **commands (Simplified computer commands.)**

Question 39

Question ID: BK05_CH05_FIB_039

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Computing power doubled continuously while hardware costs steadily _____.

Answer Key: **decreased (Hardware costs steadily decreased.)**

Question 40

Question ID: BK05_CH05_FIB_040

Type: Fill in the Blanks

Difficulty: Medium

Marks: 1

Question: Chapter 05 illustrates the collaborative evolution of information _____.

Answer Key: **technology (Evolution of information technology.)**

Question 41

Question ID: BK05_CH05_FIB_041

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Binary logic gate processing replaced mechanical gear digit _____.

Answer Key: **wheels (Replaced gear digit wheels.)**

Question 42

Question ID: BK05_CH05_FIB_042

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Micro-miniaturization catalyzed the global digital economic _____.

Answer Key: **transformation (Catalyzed global economic transformation.)**

Question 43

Question ID: BK05_CH05_FIB_043

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Data processing algorithms enable complex climate modeling and bio-chemical _____.

Answer Key: **simulations (Enable complex simulations.)**

Question 44

Question ID: BK05_CH05_FIB_044

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Lovelace's pioneering Bernoulli numbers program demonstrated algorithmic _____.

Answer Key: **looping (Demonstrated algorithmic looping.)**

Question 45

Question ID: BK05_CH05_FIB_045

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Turing's universal machine concept proved single hardware can run multi-purpose _____.

Answer Key: software (Proved hardware runs software.)

Question 46

Question ID: BK05_CH05_FIB_046

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Wartime secret code decryption demonstrated electronic calculation _____.

Answer Key: superiority (Demonstrated calculation superiority.)

Question 47

Question ID: BK05_CH05_FIB_047

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Transistor integration density defines modern microprocessor performance _____.

Answer Key: benchmarks (Defines performance benchmarks.)

Question 48

Question ID: BK05_CH05_FIB_048

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Ubiquitous computing embeds intelligent microchips into everyday consumer _____.

Answer Key: appliances (Embeds microchips into appliances.)

Question 49

Question ID: BK05_CH05_FIB_049

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Historical analysis reveals computing as a multi-century interdisciplinary _____.

Answer Key: synthesis (Multi-century interdisciplinary synthesis.)

Question 50

Question ID: BK05_CH05_FIB_050

Type: Fill in the Blanks

Difficulty: Hard

Marks: 1

Question: Chapter 05 demonstrates how mathematical theory translates into physical _____.

Answer Key: innovation (Translates theory into physical innovation.)

3. True / False Statements

> **Category:** True / False | **Total:** 50 Questions | **Marks:** 1 each

Question 1

Question ID: BK05_CH05_TF_001

Type: True/False

Difficulty: Easy

Marks: 1

Statement: The computer was invented by a single person in one afternoon.

Answer Key: False — The computer was developed over many years by several brilliant minds.

Question 2

Question ID: BK05_CH05_TF_002

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Charles Babbage designed the Difference Engine in 1822.

Answer Key: True — Text confirms Charles Babbage designed the Difference Engine in 1822.

Question 3

Question ID: BK05_CH05_TF_003

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Charles Babbage completed the Analytical Engine and built thousands of them.

Answer Key: False — He could not complete it because technology at the time was not advanced enough.

Question 4

Question ID: BK05_CH05_TF_004

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Ada Lovelace is known as the world's first computer programmer.

Answer Key: True — Ada Lovelace wrote the first algorithms and is known as the first programmer.

Question 5

Question ID: BK05_CH05_TF_005

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Alan Turing built a code-breaking machine during World War II.

Answer Key: True — Text confirms Alan Turing created a machine to break secret codes.

Question 6

Question ID: BK05_CH05_TF_006

Type: True/False

Difficulty: Easy

Marks: 1

Statement: The first electronic computer was named ENIAC.

Answer Key: True — Text confirms the first electronic computer was called ENIAC.

Question 7

Question ID: BK05_CH05_TF_007

Type: True/False

Difficulty: Easy

Marks: 1

Statement: ENIAC was built in 1822 by Charles Babbage.

Answer Key: False — ENIAC was built in 1945 by John Presper Eckert and John Mauchly.

Question 8

Question ID: BK05_CH05_TF_008

Type: True/False

Difficulty: Easy

Marks: 1

Statement: ENIAC was mammoth, filling an entire room.

Answer Key: True — ENIAC was mammoth and filled an entire room.

Question 9

Question ID: BK05_CH05_TF_009

Type: True/False

Difficulty: Easy

Marks: 1

Statement: ENIAC calculated much slower than human mathematicians.

Answer Key: False — ENIAC performed calculations much faster than any human.

Question 10

Question ID: BK05_CH05_TF_010

Type: True/False

Difficulty: Easy

Marks: 1

Statement: The invention of the microchip in the 1970s made personal computers possible.

Answer Key: True — Microchips made personal computers possible in the 1970s.

Question 11

Question ID: BK05_CH05_TF_011

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** Apple, IBM, and Microsoft helped popularize personal computers.

Answer Key: **True — These companies brought computers into homes, schools, and offices.**

Question 12

Question ID: BK05_CH05_TF_012

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** Today, computers can only be found inside large government laboratories.

Answer Key: **False — Computers are everywhere—on desks, in pockets, and in watches.**

Question 13

Question ID: BK05_CH05_TF_013

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** 'Integrator' means something which joins things so they become one.

Answer Key: **True — Vocabulary definition: Integrator = Something joining things into one.**

Question 14

Question ID: BK05_CH05_TF_014

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** 'Mammoth' means extremely small or microscopic.

Answer Key: **False — Mammoth means very big.**

Question 15

Question ID: BK05_CH05_TF_015

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** Ada Lovelace worked together with mathematician Charles Babbage.

Answer Key: **True — Ada Lovelace worked closely with Charles Babbage.**

Question 16

Question ID: BK05_CH05_TF_016

Type: True/False

Difficulty: Easy

Marks: 1

****Statement**:** Alan Turing was a British scientist.

Answer Key: **True — Alan Turing was a British scientist.**

Question 17

Question ID: BK05_CH05_TF_017

Type: True/False

Difficulty: Easy

Marks: 1

Statement: John Presper Eckert and John Mauchly built ENIAC in the United States.

Answer Key: True — Built by Eckert and Mauchly in the US.

Question 18

Question ID: BK05_CH05_TF_018

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Microchips were invented in the 1910s before World War I.

Answer Key: False — Microchips were invented in the 1970s.

Question 19

Question ID: BK05_CH05_TF_019

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Babbage's Analytical Engine is considered the first concept of a modern computer.

Answer Key: True — Analytical Engine is considered the first concept of a modern computer.

Question 20

Question ID: BK05_CH05_TF_020

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Computers have become larger and more expensive over time.

Answer Key: False — Computers became smaller, faster, and more affordable.

Question 21

Question ID: BK05_CH05_TF_021

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Ada Lovelace wrote algorithms for the Analytical Engine.

Answer Key: True — She wrote instructions/algorithms for the Analytical Engine.

Question 22

Question ID: BK05_CH05_TF_022

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Alan Turing's ideas laid the foundation for modern computers.

Answer Key: True — His ideas on thinking machines laid computer science foundations.

Question 23

Question ID: BK05_CH05_TF_023

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Chapter 05 is titled 'The Invention of the Computer'.

Answer Key: True — Chapter title is 'The Invention of the Computer'.

Question 24

Question ID: BK05_CH05_TF_024

Type: True/False

Difficulty: Easy

Marks: 1

Statement: ENIAC stood for Electronic Numerical Integrator and Computer.

Answer Key: True — Full name: Electronic Numerical Integrator and Computer.

Question 25

Question ID: BK05_CH05_TF_025

Type: True/False

Difficulty: Easy

Marks: 1

Statement: Personal computers became common in homes during the 1970s and 1980s.

Answer Key: True — Microchips in 1970s enabled PCs in homes and offices.

Question 26

Question ID: BK05_CH05_TF_026

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Charles Babbage's Difference Engine was designed to process digital images.

Answer Key: False — Difference Engine was designed to perform mathematical calculations.

Question 27

Question ID: BK05_CH05_TF_027

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Ada Lovelace realized computers could process non-numerical symbols and data.

Answer Key: True — She envisioned algorithms processing music, text, and general data.

Question 28

Question ID: BK05_CH05_TF_028

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Wartime code-breaking machines established principles of programmable electronics.

Answer Key: True — Turing's code-breaking work advanced programmable computing.

Question 29

Question ID: BK05_CH05_TF_029

Type: True/False

Difficulty: Medium

Marks: 1

Statement: ENIAC used silicon microchips to process information in 1945.

Answer Key: False — ENIAC used vacuum tubes; microchips were invented decades later in the 1970s.

Question 30

Question ID: BK05_CH05_TF_030

Type: True/False

Difficulty: Medium

Marks: 1

Statement: The microchip allowed complex circuits to fit on tiny silicon wafers.

Answer Key: True — Microchips consolidated thousands of circuits onto tiny silicon wafers.

Question 31

Question ID: BK05_CH05_TF_031

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Personal computers reduced the reliance on room-sized mainframe computers.

Answer Key: True — PCs brought computing power directly to individual desks.

Question 32

Question ID: BK05_CH05_TF_032

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Babbage's designs lacked a memory unit for storing calculation steps.

Answer Key: False — The Analytical Engine design included a 'Store' (memory) and a 'Mill' (processor).

Question 33

Question ID: BK05_CH05_TF_033

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Alan Turing proposed theoretical concepts about machines that could solve problems.

Answer Key: True — His theoretical papers defined universal computing machines.

Question 34

Question ID: BK05_CH05_TF_034

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Computer algorithms are step-by-step instructions that direct software execution.

Answer Key: True — Algorithms provide step-by-step rules for software execution.

Question 35

Question ID: BK05_CH05_TF_035

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Eckert and Mauchly built ENIAC to assist in fast mathematical computations.

Answer Key: True — Built to perform high-speed numerical integration and calculation.

Question 36

Question ID: BK05_CH05_TF_036

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Microchips caused computers to become heavier and more expensive.

Answer Key: False — Microchips made computers smaller, faster, and far more affordable.

Question 37

Question ID: BK05_CH05_TF_037

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Software companies like Microsoft developed operating systems for personal computers.

Answer Key: True — Microsoft popularized operating software for personal computers.

Question 38

Question ID: BK05_CH05_TF_038

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Modern smartwatches contain more computing power than room-sized 1940s computers.

Answer Key: True — Modern microprocessors in watches exponentially surpass ENIAC's power.

Question 39

Question ID: BK05_CH05_TF_039

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Babbage's inability to build his machine proves his mathematical theory was wrong.

Answer Key: False — His theory was correct; 19th-century mechanical manufacturing was insufficient.

Question 40

Question ID: BK05_CH05_TF_040

Type: True/False

Difficulty: Medium

Marks: 1

Statement: Chapter 05 shows that technological evolution involves continuous multi-generational effort.

Answer Key: True — Traces progress across Babbage, Lovelace, Turing, Eckert, Mauchly, and microchips.

Question 41

Question ID: BK05_CH05_TF_041

Type: True/False

Difficulty: Hard

Marks: 1

Statement: Mechanical gear computing was inherently faster than electron tube computing.

Answer Key: False — Vacuum tube electrical signals moved near light speed, far exceeding gear speeds.

Question 42

Question ID: BK05_CH05_TF_042

Type: True/False

Difficulty: Hard

Marks: 1

Statement: Ada Lovelace's documentation contained the first published computer algorithm.

Answer Key: True — Her notes on the Analytical Engine contained the first published algorithm.

Question 43

Question ID: BK05_CH05_TF_043

Type: True/False

Difficulty: Hard

Marks: 1

Statement: Turing's work at Bletchley Park was kept secret for many years after WWII.

Answer Key: True — Wartime code-breaking details were classified secret for decades.

Question 44

Question ID: BK05_CH05_TF_044

Type: True/False

Difficulty: Hard

Marks: 1

Statement: ENIAC's programming was changed by rewiring cables manually rather than downloading software.

Answer Key: True — ENIAC required physical cable patchboard rewiring to change programs.

Question 45

Question ID: BK05_CH05_TF_045

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** Silicon microchips integrated transistors, resistors, and capacitors onto one chip.

Answer Key: **True — Integrated circuits consolidated all fundamental components onto one silicon die.**

Question 46

Question ID: BK05_CH05_TF_046

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** Computer miniaturization halted entirely after the 1980s.

Answer Key: **False — Miniaturization continued rapidly into smartphones, wearables, and micro-sensors.**

Question 47

Question ID: BK05_CH05_TF_047

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** The Analytical Engine's separation of 'Mill' and 'Store' mirrors CPU and RAM today.

Answer Key: **True — Babbage's 'Mill' parallels the CPU and his 'Store' parallels RAM/memory.**

Question 48

Question ID: BK05_CH05_TF_048

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** Chapter 05 highlights that technological barriers often temporarily delay correct scientific theories.

Answer Key: **True — Babbage's theory was sound but delayed by 19th-century mechanical fabrication limitations.**

Question 49

Question ID: BK05_CH05_TF_049

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** Modern personal computers operate without any electrical energy.

Answer Key: **False — All digital electronic computers require electric power.**

Question 50

Question ID: BK05_CH05_TF_050

Type: True/False

Difficulty: Hard

Marks: 1

****Statement**:** Chapter 05 connects 19th-century mathematics with 21st-century digital lifestyle for Class 5.

Answer Key: **True — Bridges historical mathematical foundations with modern daily computer use.**

4. Short Answer Questions

> **Category**: Short Answer | **Total**: 50 Questions | **Marks**: 2 each

Question 1

Question ID: BK05_CH05_SA_001

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

Question: Was the computer invented by one single person? Explain.

Answer Key: No, the computer was not invented by one person. It was developed over many years through the contributions of several brilliant mathematicians, scientists, and engineers.

Question 2

Question ID: BK05_CH05_SA_002

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

Question: Who was Charles Babbage and what machine did he design in 1822?

Answer Key: Charles Babbage was a British mathematician who designed the Difference Engine in 1822 to perform mathematical calculations.

Question 3

Question ID: BK05_CH05_SA_003

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

Question: What was the Analytical Engine and why is it historically famous?

Answer Key: The Analytical Engine was a machine planned by Charles Babbage that is considered the first concept of a modern general-purpose computer.

Question 4

Question ID: BK05_CH05_SA_004

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

Question: Why was Charles Babbage unable to complete his Analytical Engine?

Answer Key: He could not complete it because manufacturing technology in the 19th century was not advanced enough to build the complex mechanical parts.

Question 5

Question ID: BK05_CH05_SA_005

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** Who was Ada Lovelace and why is she called the world's first computer programmer?

Answer Key: Ada Lovelace was a mathematician who worked with Babbage and wrote the first set of written instructions (algorithms) for the Analytical Engine.

Question 6

Question ID: BK05_CH05_SA_006

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What important contribution did Alan Turing make during World War II?

Answer Key: Alan Turing created a machine that could break secret enemy codes, laying theoretical foundations for modern computer science.

Question 7

Question ID: BK05_CH05_SA_007

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What was ENIAC and in which year was it built?

Answer Key: ENIAC (Electronic Numerical Integrator and Computer) was the first electronic general-purpose computer, built in 1945.

Question 8

Question ID: BK05_CH05_SA_008

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** Who built the ENIAC electronic computer in the United States?

Answer Key: John Presper Eckert and John Mauchly built the ENIAC in the United States in 1945.

Question 9

Question ID: BK05_CH05_SA_009

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** Describe the physical size and calculation capability of the ENIAC computer.

Answer Key: ENIAC was mammoth, filling an entire room, but it could perform numerical calculations much faster than any human mathematician.

Question 10

Question ID: BK05_CH05_SA_010

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What major 1970s invention made personal computers possible?

Answer Key: The invention of the microchip in the 1970s made personal computers possible by shrinking complex electronic circuits.

Question 11

Question ID: BK05_CH05_SA_011

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** Name three tech companies that helped popularize personal computers in homes and offices.

Answer Key: Apple, IBM, and Microsoft helped popularize personal computers worldwide.

Question 12

Question ID: BK05_CH05_SA_012

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** Where can computers be found in daily life today?

Answer Key: Today computers are found everywhere—on our desks, in our pockets (smartphones), and even inside wristwatches.

Question 13

Question ID: BK05_CH05_SA_013

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

****Question**:** What does the word 'mammoth' mean?

Answer Key: 'Mammoth' means very big, huge, or gigantic in physical size.

Question 14

Question ID: BK05_CH05_SA_014

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

****Question**:** What does the word 'integrator' mean?

Answer Key: 'Integrator' means something that joins separate things together so they become one unified system.

Question 15

Question ID: BK05_CH05_SA_015

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

****Question**:** What is an 'algorithm'?

Answer Key: An algorithm is a step-by-step set of rules or instructions written to solve a specific problem or complete a task.

Question 16

Question ID: BK05_CH05_SA_016

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What task was Babbage's Difference Engine created to perform?

Answer Key: It was created to perform accurate mathematical calculations and compile mathematical tables.

Question 17

Question ID: BK05_CH05_SA_017

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

****Question**:** How did personal computers differ from early room-sized computers like ENIAC?

Answer Key: Personal computers were compact, affordable, and easy for individuals to use on a desk, unlike room-sized ENIAC.

Question 18

Question ID: BK05_CH05_SA_018

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

****Question**:** What role did vacuum tubes play in 1940s electronic computers?

Answer Key: Vacuum tubes acted as fast electronic switches to process electrical signals and numbers far faster than mechanical gears.

Question 19

Question ID: BK05_CH05_SA_019

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

Question: In which century did Ada Lovelace write the first computer program?

Answer Key: She wrote the first computer program in the 19th century (1840s).

Question 20

Question ID: BK05_CH05_SA_020

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

Question: What does the acronym ENIAC stand for?

Answer Key: ENIAC stands for Electronic Numerical Integrator and Computer.

Question 21

Question ID: BK05_CH05_SA_021

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

Question: How did computers change regarding speed, size, and cost over time?

Answer Key: Computers became drastically smaller in size, much faster in calculation speed, and far more affordable in cost.

Question 22

Question ID: BK05_CH05_SA_022

Type: Short Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 2

Question: Why did Alan Turing's work help create 'thinking machines'?

Answer Key: He conceptualized machines that could follow logical rules to solve problems autonomously, establishing artificial intelligence concepts.

Question 23

Question ID: BK05_CH05_SA_023

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

Question: What contribution did IBM make to computer history?

Answer Key: IBM helped manufacture and popularize business and personal computers for offices and schools around the world.

Question 24

Question ID: BK05_CH05_SA_024

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What contribution did Microsoft make to computer history?

Answer Key: Microsoft developed user-friendly software and operating systems that allowed ordinary people to operate personal computers.

Question 25

Question ID: BK05_CH05_SA_025

Type: Short Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 2

****Question**:** What title is given to Chapter 05?

Answer Key: The title of Chapter 05 is 'The Invention of the Computer'.

Question 26

Question ID: BK05_CH05_SA_026

Type: Short Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 2

****Question**:** Explain why Charles Babbage's Analytical Engine is considered the true precursor to modern computers.

Answer Key: It introduced the core architecture of modern computing: input, a processing unit ('Mill'), memory storage ('Store'), and output mechanisms, separating hardware from software logic.

Question 27

Question ID: BK05_CH05_SA_027

Type: Short Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 2

****Question**:** How did Ada Lovelace's vision go beyond Charles Babbage's original mathematical focus?

Answer Key: Babbage focused on numerical math, whereas Lovelace realized the machine could process any symbols or data (including music and graphics) if represented algorithmically.

Question 28

Question ID: BK05_CH05_SA_028

Type: Short Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 2

****Question**:** Describe the historical significance of Alan Turing's code-breaking machine during World War II.

Answer Key: His machine automated the decryption of complex military ciphers, saving millions of lives and proving that electronic machines could execute complex logical reasoning.

Question 29

Question ID: BK05_CH05_SA_029

Type: Short Answer

Difficulty: Medium

Bloom Level: Comparing

Marks: 2

****Question**:** Contrast the mechanical gears of Babbage's engines with the electronic circuits of ENIAC.

Answer Key: Babbage's engines relied on physical metal wheels turning mechanically, which was slow and prone to wear. ENIAC used electronic vacuum tubes operating at light speed.

Question 30

Question ID: BK05_CH05_SA_030

Type: Short Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 2

****Question**:** Why was the invention of the microchip in the 1970s a turning point for global society?

Answer Key: It compressed millions of electronic components onto a small silicon chip, making computing powerful, portable, and cheap enough for individual ownership.

Question 31

Question ID: BK05_CH05_SA_031

Type: Short Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 2

****Question**:** How did companies like Apple and Microsoft change the way people interact with technology?

Answer Key: They created graphical interfaces, personal computer hardware, and intuitive software, turning technical industrial machines into friendly consumer appliances.

Question 32

Question ID: BK05_CH05_SA_032

Type: Short Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 2

****Question**:** Why did Babbage's failure to complete the Analytical Engine not diminish his legacy?

Answer Key: Because his written designs and logical concepts were mathematically sound and served as the architectural blueprint when electronics developed a century later.

Question 33

Question ID: BK05_CH05_SA_033

Type: Short Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 2

****Question**:** Describe how computers evolved from single-purpose tools to general-purpose devices.

Answer Key: Early tools performed single tasks (math or code-breaking). General-purpose computers can run unlimited different software programs, performing tasks from word processing to gaming.

Question 34

Question ID: BK05_CH05_SA_034

Type: Short Answer

Difficulty: Medium

Bloom Level: Applying

Marks: 2

****Question**:** How do algorithms govern the operation of modern smartphone applications?

Answer Key: Algorithms process user inputs (touches, swipes), make logical calculations, retrieve data from servers, and display visual results on screen step-by-step.

Question 35

Question ID: BK05_CH05_SA_035

Type: Short Answer

Difficulty: Medium

Bloom Level: Understanding

Marks: 2

****Question**:** Summarize Chapter 05 in four concise sentences.

Answer Key: The computer developed over centuries through contributions from several brilliant pioneers. Charles Babbage designed early calculation engines, while Ada Lovelace wrote the first algorithms. Alan Turing created wartime code-breaking machines, and ENIAC emerged as the first room-sized electronic computer in 1945. The 1970s microchip enabled Apple, IBM, and Microsoft to make personal computers ubiquitous worldwide.

Question 36

Question ID: BK05_CH05_SA_036

Type: Short Answer

Difficulty: Medium

Bloom Level: Understanding

Marks: 2

****Question**:** Why were 1940s room-sized computers unsuitable for home or school use?

Answer Key: They were immense (filling entire rooms), consumed huge electric power, generated intense heat, required team maintenance, and cost millions of dollars.

Question 37

Question ID: BK05_CH05_SA_037

Type: Short Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 2

****Question**:** How did Ada Lovelace demonstrate that women played a vital role in early STEM history?

Answer Key: At a time when female scientists were rare, her brilliant mathematical insights created computer programming, establishing women as founding pioneers of software engineering.

Question 38

Question ID: BK05_CH05_SA_038

Type: Short Answer

Difficulty: Medium

Bloom Level: Comparing

Marks: 2

****Question**:** What is the relationship between computer hardware and computer software?

Answer Key: Hardware represents physical circuits, chips, and screens. Software represents written algorithmic instructions that tell the physical hardware what tasks to perform.

Question 39

Question ID: BK05_CH05_SA_039

Type: Short Answer

Difficulty: Medium

Bloom Level: Applying

Marks: 2

****Question**:** How have modern computers transformed the way students learn in schools?

Answer Key: Computers provide instant access to digital libraries, interactive educational software, online research, global communication, and remote learning platforms.

Question 40

Question ID: BK05_CH05_SA_040

Type: Short Answer

Difficulty: Medium

Bloom Level: Applying

Marks: 2

****Question**:** What advice would you give to a Class 5 student interested in learning computer programming?

Answer Key: Start by learning basic logical step-by-step thinking, practice block-based coding, stay curious about how algorithms work, and build fun problem-solving projects.

Question 41

Question ID: BK05_CH05_SA_041

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 2

****Question**:** Critique the physical limitations of ENIAC's 18,000 vacuum tubes.

Answer Key: Vacuum tubes burned out frequently, generated massive heat, consumed 150 kW of power, and required manual cable patchboard rewiring to change programs, making maintenance difficult.

Question 42

Question ID: BK05_CH05_SA_042

Type: Short Answer

Difficulty: Hard

Bloom Level: Analyzing

Marks: 2

****Question**:** Deconstruct the transition from mechanical decimal computing to digital binary computing.

Answer Key: Decimal mechanics required ten-position gears for digits 0-9. Digital binary uses two electrical voltage states (ON = 1, OFF = 0), enabling ultra-fast, reliable logic gates.

Question 43

Question ID: BK05_CH05_SA_043

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 2

****Question**:** Evaluate the impact of widespread computer access on global human collaboration.

Answer Key: Universal computing connects global researchers, businesses, and communities instantly, allowing collaborative problem-solving across time zones and geographic borders.

Question 44

Question ID: BK05_CH05_SA_044

Type: Short Answer

Difficulty: Hard

Bloom Level: Comparing

Marks: 2

****Question**:** Compare Charles Babbage's 1822 Difference Engine with John Presper Eckert's 1945 ENIAC.

Answer Key: Difference Engine: Mechanical, gear-driven, single-task math calculator. ENIAC: Electronic, vacuum tube-driven, high-speed multi-task numerical integrator.

Question 45

Question ID: BK05_CH05_SA_045

Type: Short Answer

Difficulty: Hard

Bloom Level: Creating

Marks: 2

****Question**:** Formulate a short story about a child visiting the room-sized ENIAC computer in 1945.

Answer Key: 'Ten-year-old Thomas stared in awe at tall metal cabinets filling the vast room. Thousands of glass vacuum tubes glowed red as heavy cables hummed, calculating artillery tables faster than fifty human mathematicians.'

Question 46

Question ID: BK05_CH05_SA_046

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 2

****Question**:** Assess the importance of open theoretical publishing in computer science history.

Answer Key: Babbage and Lovelace published detailed notes that survived a century, allowing post-WWII scientists to revive and realize their visionary computing architecture.

Question 47

Question ID: BK05_CH05_SA_047

Type: Short Answer

Difficulty: Hard

Bloom Level: Analyzing

Marks: 2

****Question**:** Analyze how software user interfaces (GUIs) democratized technology access.

Answer Key: GUIs replaced cryptic text command lines with visual icons, folders, and mouse pointers, allowing people of all ages and education levels to operate computers easily.

Question 48

Question ID: BK05_CH05_SA_048

Type: Short Answer

Difficulty: Hard

Bloom Level: Synthesizing

Marks: 2

****Question**:** Synthesize how Chapter 05 integrates mathematics, engineering, and history.

Answer Key: Combines 19th-century mathematics (Babbage/Lovelace), 20th-century history (WWII/1970s), and engineering breakthroughs (vacuum tubes/microchips).

Question 49

Question ID: BK05_CH05_SA_049

Type: Short Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 2

****Question**:** Critique the claim: 'Computing technology has reached its ultimate limit today.'

Answer Key: False; emerging fields like quantum computing, optical computing, and neural processors continue to expand processing power exponentially beyond current silicon limits.

Question 50

Question ID: BK05_CH05_SA_050

Type: Short Answer

Difficulty: Hard

Bloom Level: Creating

Marks: 2

****Question****: Formulate a 4-line stanza summarizing the history of computing.

Answer Key: 'Babbage planned engines of gears and steel,

Lovelace wrote code to make logic real;

Turing and ENIAC brought electronic light,

Now microchips put the world in our sight!'

5. Long Answer & HOTS Questions

> **Category**: Long Answer | **Total**: 50 Questions | **Marks**: 5 each

Question 1

Question ID: BK05_CH05_LA_001

Type: Long Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 5

Question: Describe the early history of computing, detailing the contributions of Charles Babbage and Ada Lovelace in the 19th century.

Answer Key: The journey of computer invention began in the early 19th century with British mathematician Charles Babbage. In 1822, Babbage designed the Difference Engine to perform mathematical calculations automatically. Later, he planned an advanced machine called the Analytical Engine, which included a processing unit, memory storage, and input mechanisms—making it the first true concept of a modern computer. However, Babbage could not complete building it because 19th-century manufacturing technology was not precise enough. Working alongside Babbage, Ada Lovelace studied the Analytical Engine and wrote the first set of written instructions, or algorithms, for it. Because of her visionary work, Ada Lovelace is celebrated as the world's first computer programmer.

Question 2

Question ID: BK05_CH05_LA_002

Type: Long Answer

Difficulty: Easy

Bloom Level: Remembering

Marks: 5

Question: Explain the contributions of Alan Turing, John Presper Eckert, and John Mauchly to 20th-century electronic computing.

Answer Key: In the 20th century, computing transitioned from mechanical designs to practical electronic systems. During World War II, British scientist Alan Turing built a landmark machine that could break secret enemy military codes. His theoretical papers on machines that could follow logical rules laid the foundations for computer science and artificial intelligence. In 1945, American engineers John Presper Eckert and John Mauchly built ENIAC (Electronic Numerical Integrator and Computer), recognized as the first general-purpose electronic computer. Although ENIAC was mammoth and filled an entire room, its electronic vacuum tubes allowed it to calculate numerical problems infinitely faster than human mathematicians.

Question 3

Question ID: BK05_CH05_LA_003

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

Question: Describe how the 1970s microchip revolution and companies like Apple, IBM, and Microsoft brought computers into everyday life.

Answer Key: For decades after ENIAC, computers remained room-sized mainframes used primarily by military and research institutions. The turning point arrived in the 1970s with the invention of the microchip, which compressed thousands of electronic circuits onto a tiny silicon wafer. Microchips enabled engineers to build compact, affordable 'personal computers' (PCs) suitable for individual desks. Tech companies played key roles in mass adoption: IBM manufactured reliable personal computers for corporate offices and schools, while Apple and Microsoft created user-friendly hardware and operating software. Together, these innovations brought computers into homes, classrooms, and businesses worldwide.

Question 4

Question ID: BK05_CH05_LA_004

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** Trace the physical evolution of computers regarding size, speed, cost, and everyday accessibility from 1822 to today.

Answer Key: Computer hardware underwent an astounding physical evolution over two centuries:

1. ****1822 (Babbage Era)**:** Heavy mechanical gear engines built from brass and iron, slow and incomplete.
2. ****1945 (ENIAC Era)**:** Room-sized mammoth electronic mainframes weighing 30 tons with 18,000 glowing vacuum tubes.
3. ****1970s (Microchip Era)**:** Desktop personal computers enabled by silicon chips, fitting on study tables in homes and offices.
4. ****Today (Modern Era)**:** Ultra-compact, inexpensive microprocessors embedded everywhere—on desks, in pocket smartphones, and inside wristwatches.

Question 5

Question ID: BK05_CH05_LA_005

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** Explain the vocabulary words from Chapter 05: Integrator, Mammoth, and Algorithm with definitions and example sentences.

Answer Key: 1. **Integrator:** Something which joins separate parts together into a unified system.

***Sentence*:** ENIAC stood for Electronic Numerical Integrator and Computer.

2. ****Mammoth**:** Extremely big or huge in physical size. ***Sentence*:** The ENIAC computer was mammoth and filled an entire room.
3. ****Algorithm**:** A step-by-step set of written rules or instructions to solve a problem. ***Sentence*:** Ada Lovelace wrote the world's first computer algorithm.

Question 6

Question ID: BK05_CH05_LA_006

Type: Long Answer

Difficulty: Easy

Bloom Level: Evaluating

Marks: 5

****Question**:** Discuss why Ada Lovelace's work remains historically significant in computer science.

Answer Key: Ada Lovelace's work is historically significant because she foresaw that computing machines were capable of more than mere arithmetic. While Charles Babbage viewed his engine as a high-speed calculator, Lovelace understood that if a machine could manipulate numbers, those numbers could represent music notes, letters, or symbols. By writing an algorithm to calculate Bernoulli numbers, she proved that software algorithms could direct hardware tasks, founding the discipline of computer programming.

Question 7

Question ID: BK05_CH05_LA_007

Type: Long Answer

Difficulty: Easy

Bloom Level: Analyzing

Marks: 5

****Question**:** How did World War II accelerate computer science breakthroughs as described in Chapter 05?

Answer Key: World War II created an urgent military necessity for rapid calculation and code-breaking. Manual encryption and ballistics math were too slow for warfare. This pressure drove governments to fund groundbreaking research, leading Alan Turing to build electromechanical code-breaking machines at Bletchley Park and Eckert and Mauchly to construct ENIAC. These wartime projects proved the speed of electronic computing, spurring the postwar commercial computer industry.

Question 8

Question ID: BK05_CH05_LA_008

Type: Long Answer

Difficulty: Easy

Bloom Level: Comparing

Marks: 5

****Question**:** Explain the difference between Charles Babbage's Difference Engine and his Analytical Engine.

Answer Key: The Difference Engine (designed in 1822) was a specialized mechanical calculator built specifically to compute mathematical tables using polynomial addition. The Analytical Engine (planned later) was a general-purpose programmable machine. It included a central processing unit ('Mill'), memory storage ('Store'), punch-card data inputs, and sequential control—making it the true theoretical conceptual ancestor of modern computers.

Question 9

Question ID: BK05_CH05_LA_009

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** Summarize Chapter 05 in five detailed bullet points.

Answer Key: - The computer was developed over centuries by multiple brilliant mathematicians and scientists.

- Charles Babbage designed the Difference Engine (1822) and Analytical Engine, while Ada Lovelace wrote the first algorithms.

- Alan Turing created World War II code-breaking machines, establishing computer science foundations.
- ENIAC, built in 1945 by Eckert and Mauchly, was the first mammoth room-sized electronic computer.
- The 1970s microchip enabled Apple, IBM, and Microsoft to make personal computers affordable and ubiquitous worldwide.

Question 10

Question ID: BK05_CH05_LA_010

Type: Long Answer

Difficulty: Easy

Bloom Level: Applying

Marks: 5

Question: How do computers help humans in work, learning, and communication today?

Answer Key: Computers transform modern human life across three main areas:

1. **Work**: Automating financial accounting, designing architecture, managing global supply chains, and storing data.
2. **Learning**: Providing digital textbooks, educational research tools, interactive simulations, and distance learning platforms.
3. **Communication**: Powering email, video conferencing, social networks, and instant messaging across global distances.

Question 11

Question ID: BK05_CH05_LA_011

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

Question: Why could Charles Babbage not complete the Analytical Engine during his lifetime?

Answer Key: Babbage faced 19th-century engineering constraints. Metal machining was done by hand, making it impossible to produce thousands of identical cogwheels without friction errors. Additionally, government funding was withdrawn, leaving his masterwork unbuilt.

Question 12

Question ID: BK05_CH05_LA_012

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

Question: Explain how vacuum tubes allowed ENIAC to calculate faster than mechanical gears.

Answer Key: Vacuum tubes have no moving physical parts. They manipulate streams of electrons at near-light speeds using electrical voltage, allowing ENIAC to perform 5,000 additions per second compared to a few additions per second on mechanical gears.

Question 13

Question ID: BK05_CH05_LA_013

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** Describe the role of microchips in shrinking computer hardware.

Answer Key: Microchips (integrated circuits) replace bulky wires, vacuum tubes, and separate transistors by printing millions of microscopic electronic components directly onto a tiny silicon wafer, dramatically reducing size and power consumption.

Question 14

Question ID: BK05_CH05_LA_014

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** How did Apple, IBM, and Microsoft contribute to the personal computer revolution?

Answer Key: IBM established standardized business PC hardware; Apple introduced intuitive graphical user interfaces and aesthetic design; Microsoft authored widespread operating software (MS-DOS/Windows) that made PCs easy to operate.

Question 15

Question ID: BK05_CH05_LA_015

Type: Long Answer

Difficulty: Easy

Bloom Level: Comparing

Marks: 5

****Question**:** Compare the performance of ENIAC in 1945 with a modern smartphone.

Answer Key: ENIAC weighed 30 tons, filled a room, consumed 150 kW of power, and did 5,000 operations/sec. A smartphone weighs a few ounces, fits in a pocket, runs on a tiny battery, and performs billions of operations/sec.

Question 16

Question ID: BK05_CH05_LA_016

Type: Long Answer

Difficulty: Easy

Bloom Level: Evaluating

Marks: 5

****Question**:** Why is computer software just as important as computer hardware?

Answer Key: Hardware is merely physical machinery. Without software algorithms to direct signal paths and calculation steps, hardware cannot perform any useful tasks. Software gives hardware functional purpose.

Question 17

Question ID: BK05_CH05_LA_017

Type: Long Answer

Difficulty: Easy

Bloom Level: Evaluating

Marks: 5

****Question**:** How did Alan Turing's code-breaking machine help shorten World War II?

Answer Key: By automating the decryption of secret enemy military ciphers, Turing's machine provided vital intelligence about troop movements and naval plans, saving millions of lives and shortening the war.

Question 18

Question ID: BK05_CH05_LA_018

Type: Long Answer

Difficulty: Easy

Bloom Level: Applying

Marks: 5

****Question**:** Explain the concept of an 'algorithm' using a daily life example.

Answer Key: An algorithm is a step-by-step recipe. For example, a recipe for making tea (boil water → add tea leaves → pour milk → strain) is an algorithm that guarantees the same result when steps are followed in order.

Question 19

Question ID: BK05_CH05_LA_019

Type: Long Answer

Difficulty: Easy

Bloom Level: Evaluating

Marks: 5

****Question**:** How does Chapter 05 illustrate the value of persistence in scientific research?

Answer Key: Despite Babbage not seeing his engine built and Lovelace's notes being forgotten for decades, their theoretical persistence created ideas that eventually reshaped global civilization a century later.

Question 20

Question ID: BK05_CH05_LA_020

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** Discuss how computers are embedded in modern vehicles and household appliances.

Answer Key: Modern cars, micro-ovens, washing machines, and smart TVs contain dedicated microcontrollers (tiny computers) that automatically regulate engine fuel, washing cycles, temperature, and video display.

Question 21

Question ID: BK05_CH05_LA_021

Type: Long Answer

Difficulty: Easy

Bloom Level: Creating

Marks: 5

****Question**:** Re-write the story of computing from the perspective of Ada Lovelace in 1843.

Answer Key: 'As I looked at Babbage's drawing of the Analytical Engine, I saw more than numbers. I saw that by weaving algorithms like a Jacquard loom weaves silk, this machine could process words, music, and thought itself!'

Question 22

Question ID: BK05_CH05_LA_022

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** What role does binary code (1s and 0s) play in modern digital computing?

Answer Key: Binary code converts all data—text, images, sound, and video—into combinations of 1s (voltage ON) and 0s (voltage OFF), which microchips process at lightning speed through logic gates.

Question 23

Question ID: BK05_CH05_LA_023

Type: Long Answer

Difficulty: Easy

Bloom Level: Analyzing

Marks: 5

****Question**:** How did personal computers empower small businesses and independent workers?

Answer Key: PCs allowed small businesses to handle accounting, graphic design, document printing, and customer records in-house without hiring expensive corporate computing services.

Question 24

Question ID: BK05_CH05_LA_024

Type: Long Answer

Difficulty: Easy

Bloom Level: Analyzing

Marks: 5

****Question**:** Analyze why Chapter 05 is titled 'The Invention of the Computer' rather than 'The Inventor of the Computer'.

Answer Key: The title uses 'Invention' to reflect that computing was not created by a single inventor, but evolved as a multi-stage collective achievement across centuries.

Question 25

Question ID: BK05_CH05_LA_025

Type: Long Answer

Difficulty: Easy

Bloom Level: Understanding

Marks: 5

****Question**:** What future fields of study are emerging from computer science today?

Answer Key: Emerging fields include Artificial Intelligence (AI), Quantum Computing, Cybersecurity, Robotics, Data Science, and Bio-informatics.

Question 26

Question ID: BK05_CH05_LA_026

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

****Question**:** Critically analyze how the separation of hardware and software enabled the software industry.

Answer Key: When early computers (ENIAC) required physical rewiring for each new problem, software did not exist independently. The separation of hardware (central processing units and memory) from software (stored program algorithms) allowed computer code to be written, copied, distributed, and updated without altering physical circuits. This conceptual breakthrough gave birth to the global software industry led by companies like Microsoft.

Question 27

Question ID: BK05_CH05_LA_027

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

****Question**:** Examine the technological shift from mechanical gears (Babbage) to silicon microchips (1970s).

Answer Key: The technological shift spans three mechanical-to-digital physics leaps:

1. ****Mechanical Gears**:** Heavy, friction-bound, slow moving parts subject to mechanical wear and manufacturing tolerances.
2. ****Vacuum Tubes**:** Frictionless electron streams in glass bulbs operating at kilohertz speeds, but fragile and heat-producing.
3. ****Silicon Microchips**:** Solid-state micro-electronics integrating billions of microscopic transistors onto silicon dies, operating silently at gigahertz speeds.

Question 28

Question ID: BK05_CH05_LA_028

Type: Long Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 5

****Question**:** Evaluate the role of diversity and female leadership in computer history through Ada Lovelace.

Answer Key: Ada Lovelace's foundational role demonstrates that computer science was co-created by female visionaries. At a time when women were excluded from formal scientific academies, Lovelace's intellectual partnership with Babbage produced the world's first published program. Celebrating her legacy inspires gender diversity in modern STEM disciplines.

Question 29

Question ID: BK05_CH05_LA_029

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

****Question**:** Discuss how the miniaturization of computers changed human social behavior.

Answer Key: Miniaturization transformed computers from isolated institutional mainframes into personal companions. Carrying computers in pockets (smartphones) enables instant social networking, continuous

navigation, immediate information retrieval, and remote work, blending digital technology seamlessly into daily human behavior.

Question 30

Question ID: BK05_CH05_LA_030

Type: Long Answer

Difficulty: Medium

Bloom Level: Creating

Marks: 5

****Question**:** Design an interactive classroom debate topic based on Chapter 05.

Answer Key: Debate Topic: 'Has the ubiquity of personal computers made human life better or more complicated?'

- ****Side A (Pro)**:** Computers automate tedious labor, connect global families, provide instant education, and advance medicine.

- ****Side B (Con)**:** Computers create digital addiction, cause data privacy risks, reduce physical activity, and blur work-life boundaries.

Question 31

Question ID: BK05_CH05_LA_031

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

****Question**:** How did Alan Turing's theoretical work resolve the problem of universal computation?

Answer Key: Turing proved mathematically that a single machine ('Universal Turing Machine') could execute any computable task simply by reading different program instructions from tape, eliminating the need for dedicated single-purpose hardware.

Question 32

Question ID: BK05_CH05_LA_032

Type: Long Answer

Difficulty: Medium

Bloom Level: Comparing

Marks: 5

****Question**:** Contrast the commercial strategy of IBM with the commercial strategy of Apple in the 1970s/1980s.

Answer Key: IBM targeted corporate businesses with standardized, open-architecture desktop PCs. Apple targeted creative individuals and schools with integrated, visually elegant, user-friendly hardware and software ecosystems.

Question 33

Question ID: BK05_CH05_LA_033

Type: Long Answer

Difficulty: Medium

Bloom Level: Understanding

Marks: 5

****Question**:** Why was the invention of ENIAC kept confidential during its early development?

Answer Key: ENIAC was developed during World War II under United States military sponsorship to calculate secret artillery firing tables, keeping design details classified until after the war.

Question 34

Question ID: BK05_CH05_LA_034

Type: Long Answer

Difficulty: Medium

Bloom Level: Applying

Marks: 5

****Question**:** Explain how computer algorithms enable modern search engines to find information.

Answer Key: Search engine algorithms index billions of web pages, analyze search keywords in milliseconds, score relevancy using mathematical rules, and display the best answers instantly.

Question 35

Question ID: BK05_CH05_LA_035

Type: Long Answer

Difficulty: Medium

Bloom Level: Understanding

Marks: 5

****Question**:** How did Charles Babbage's use of punch cards originate from the textile industry?

Answer Key: Babbage adapted punch cards from Joseph Marie Jacquard's automated weaving loom, which used hole-patterned cards to direct complex fabric patterns automatically.

Question 36

Question ID: BK05_CH05_LA_036

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

****Question**:** Analyze why personal computers succeeded in the 1970s whereas earlier microcomputers failed.

Answer Key: The 1970s brought affordable silicon microchips, standardized monitors, floppy disk storage, and accessible programming languages (BASIC), creating complete, practical systems.

Question 37

Question ID: BK05_CH05_LA_037

Type: Long Answer

Difficulty: Medium

Bloom Level: Evaluating

Marks: 5

****Question**:** Evaluate the importance of error-checking in early computing machinery.

Answer Key: Mechanical and vacuum tube systems frequently suffered calculation errors from gear slips or burnt tubes. Inventors designed internal mathematical checks to ensure reliable results.

Question 38

Question ID: BK05_CH05_LA_038

Type: Long Answer

Difficulty: Medium

Bloom Level: Understanding

Marks: 5

Question: How do modern operating systems manage computer hardware resources?

Answer Key: Operating systems act as software managers, allocating CPU processing time, RAM memory space, storage access, and peripheral device connections smoothly between running programs.

Question 39

Question ID: BK05_CH05_LA_039

Type: Long Answer

Difficulty: Medium

Bloom Level: Analyzing

Marks: 5

Question: Describe the impact of computer development on space exploration.

Answer Key: Computers processed complex orbital trajectory math, controlled spacecraft guidance systems in real time, and analyzed satellite imaging, enabling moon landings and deep space probes.

Question 40

Question ID: BK05_CH05_LA_040

Type: Long Answer

Difficulty: Medium

Bloom Level: Creating

Marks: 5

Question: Construct a fictional speech by Charles Babbage visiting a modern smartphone store.

Answer Key: 'I am astonished! In this palm-sized glass wafer lies an Analytical Engine billions of times faster than my brass cogs, running programs my dear Ada and I could only dream of!'

Question 41

Question ID: BK05_CH05_LA_041

Type: Long Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 5

Question: Critique the structural limitations of the von Neumann computer architecture.

Answer Key: The von Neumann architecture (separating CPU and memory via a shared bus) creates the 'von Neumann bottleneck'. Physical data transfer speed between memory and processor limits overall throughput. Modern hardware addresses this through multi-core CPUs, multi-level high-speed cache memory, and parallel processing.

Question 42

Question ID: BK05_CH05_LA_042

Type: Long Answer

Difficulty: Hard

Bloom Level: Analyzing

Marks: 5

****Question**:** Deconstruct the mathematical concept of 'computability' pioneered by Alan Turing.

Answer Key: Turing defined computability by proving which mathematical problems can be solved step-by-step by an automated machine and which cannot (the Halting Problem). This established the formal theoretical boundaries of computer science.

Question 43

Question ID: BK05_CH05_LA_043

Type: Long Answer

Difficulty: Hard

Bloom Level: Synthesizing

Marks: 5

****Question**:** Synthesize the historical arc of computing across three centuries.

Answer Key: 19th Century: Mechanical theory & mathematical algorithms (Babbage/Lovelace).

20th Century: Electronic vacuum tubes, code-breaking, mainframes, & microchips (Turing/ENIAC/Apple/IBM/Microsoft).

21st Century: Ubiquitous mobile, cloud, internet-of-things, & artificial intelligence computing.

Question 44

Question ID: BK05_CH05_LA_044

Type: Long Answer

Difficulty: Hard

Bloom Level: Creating

Marks: 5

****Question**:** Formulate an advanced assessment task for Class 5 students comparing TV and computer history.

Answer Key: Task: Create a comparative timeline showing how television (Baird/Farnsworth) and computer (Babbage/Turing/ENIAC) technologies evolved in parallel during the 20th century to converge into modern smart devices.

Question 45

Question ID: BK05_CH05_LA_045

Type: Long Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 5

****Question**:** Evaluate the ecological and environmental impact of electronic computing hardware.

Answer Key: While computers boost environmental research, electronic waste (e-waste), rare earth metal mining, and high data center energy consumption pose significant ecological challenges requiring sustainable recycling.

Question 46

Question ID: BK05_CH05_LA_046

Type: Long Answer

Difficulty: Hard

Bloom Level: Comparing

Marks: 5

****Question**:** Compare ENIAC's patchboard programming with modern high-level languages like Python.

Answer Key: ENIAC required manual physical cable plugging and switch flipping to change logic. Python uses human-readable text code translated automatically into machine binary by compilers.

Question 47

Question ID: BK05_CH05_LA_047

Type: Long Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 5

****Question**:** Discuss the philosophical implications of artificial intelligence as envisioned by Alan Turing.

Answer Key: Turing posed the question 'Can machines think?', suggesting that if a machine behaves indistinguishably from a human in conversation, it exhibits artificial intelligence, sparking debate on consciousness.

Question 48

Question ID: BK05_CH05_LA_048

Type: Long Answer

Difficulty: Hard

Bloom Level: Analyzing

Marks: 5

****Question**:** Analyze how cloud computing alters traditional personal computer storage.

Answer Key: Cloud computing shifts data storage and heavy processing from local hard drives to remote data center servers connected via internet, allowing access from any device anywhere.

Question 49

Question ID: BK05_CH05_LA_049

Type: Long Answer

Difficulty: Hard

Bloom Level: Evaluating

Marks: 5

****Question**:** Draft an analytical commentary on the line: 'The computer, one of the most important inventions in the world, has a fascinating history.'

Answer Key: This opening sentence sets a tone of historical awe. It frames the computer not as a static consumer product, but as a dynamic, evolving human achievement shaping global civilization.

Question 50

Question ID: BK05_CH05_LA_050

Type: Long Answer

Difficulty: Hard

Bloom Level: Synthesizing

Marks: 5

****Question**:** Synthesize the ultimate educational value of Chapter 05 for primary school computer literacy.

Answer Key: Chapter 05 teaches that computing is a human story of persistent problem-solving, intellectual collaboration, and continuous technological refinement, inspiring students to become creators of technology.

6. Extract Based Questions

> **Category**: Extract Based | **Total**: 10 Extract Sets (50 Questions) | **Marks**: 1 per sub-question

Extract Set 1

> **"The computer, one of the most important inventions in the world, has a fascinating history. It was not invented by one person but was developed over many years by several brilliant minds."**

Question 1

Question ID: BK05_CH05_EXT_001

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What is described as one of the most important inventions in the world?

Answer Key: The computer.

Question 2

Question ID: BK05_CH05_EXT_002

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Was the computer invented by a single person?

Answer Key: No, it was developed over many years by several brilliant minds.

Question 3

Question ID: BK05_CH05_EXT_003

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: How is the history of the computer characterized?

Answer Key: It has a fascinating history.

Question 4

Question ID: BK05_CH05_EXT_004

Type: Extract Based

Difficulty: Medium

Bloom Level: Understanding

Marks: 1

Question: Why did computer invention require multiple pioneers?

Answer Key: Because it involved complex stages: mechanical design, algorithmic logic, electronic circuits, and microchips.

Question 5

Question ID: BK05_CH05_EXT_005

Type: Extract Based

Difficulty: Medium

Bloom Level: Evaluating

Marks: 1

****Question**:** What lesson does this extract convey about major technological breakthroughs?

Answer Key: Major breakthroughs result from multi-generational collaboration and continuous innovation.

Extract Set 2

> **"The journey began in the early 19th century with a mathematician named Charles Babbage. He designed a machine called the Difference Engine in 1822, which could perform mathematical calculations."**

Question 6

Question ID: BK05_CH05_EXT_006

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** When did the journey of computer invention begin?

Answer Key: In the early 19th century.

Question 7

Question ID: BK05_CH05_EXT_007

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who was Charles Babbage?

Answer Key: A British mathematician.

Question 8

Question ID: BK05_CH05_EXT_008

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** What machine did Charles Babbage design in 1822?

Answer Key: The Difference Engine.

Question 9

Question ID: BK05_CH05_EXT_009

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What task was the Difference Engine created to perform?

Answer Key: Mathematical calculations.

Question 10

Question ID: BK05_CH05_EXT_010

Type: Extract Based

Difficulty: Easy

Bloom Level: Understanding

Marks: 1

Question: What does the word 'calculations' mean?

Answer Key: Mathematical processes of counting or computing numbers.

Extract Set 3

> "Later, he planned an even more advanced machine called the Analytical Engine, considered the first concept of a modern computer. However, Babbage could not complete it because technology at the time was not advanced enough."

Question 11

Question ID: BK05_CH05_EXT_011

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What advanced machine did Babbage plan after the Difference Engine?

Answer Key: The Analytical Engine.

Question 12

Question ID: BK05_CH05_EXT_012

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What is the Analytical Engine considered in computer history?

Answer Key: The first concept of a modern computer.

Question 13

Question ID: BK05_CH05_EXT_013

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Why could Babbage not complete the Analytical Engine?

Answer Key: Because technology at the time was not advanced enough.

Question 14

Question ID: BK05_CH05_EXT_014

Type: Extract Based

Difficulty: Medium

Bloom Level: Understanding

Marks: 1

****Question**:** What engineering capability was lacking in the 19th century?

Answer Key: Precision metal machining to build thousands of delicate gear parts.

Question 15

Question ID: BK05_CH05_EXT_015

Type: Extract Based

Difficulty: Medium

Bloom Level: Evaluating

Marks: 1

****Question**:** Why is Babbage celebrated despite not completing his machine?

Answer Key: Because his conceptual architecture established the theoretical blueprint for modern computers.

Extract Set 4

> **"**Another important figure was Ada Lovelace, who worked with Babbage. She wrote first instructions, or algorithms, for his Analytical Engine. This is why she is known as the world's first computer programmer."*****

Question 16

Question ID: BK05_CH05_EXT_016

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who worked closely with Charles Babbage?

Answer Key: Ada Lovelace.

Question 17

Question ID: BK05_CH05_EXT_017

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** What did Ada Lovelace write for the Analytical Engine?

Answer Key: The first computer instructions, or algorithms.

Question 18

Question ID: BK05_CH05_EXT_018

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What title is Ada Lovelace known by in computer history?

Answer Key: The world's first computer programmer.

Question 19

Question ID: BK05_CH05_EXT_019

Type: Extract Based

Difficulty: Easy

Bloom Level: Understanding

Marks: 1

Question: What is an 'algorithm'?

Answer Key: A step-by-step set of instructions for solving a problem.

Question 20

Question ID: BK05_CH05_EXT_020

Type: Extract Based

Difficulty: Medium

Bloom Level: Analyzing

Marks: 1

Question: Why was Lovelace's achievement visionary?

Answer Key: She foresaw that machines could process non-numerical data like music and graphics using algorithmic code.

Extract Set 5

> "In the 20th century, computers became more practical. During World War II, Alan Turing, a British scientist, created a machine that could break secret codes. His ideas about machines that could think and solve problems laid the foundation for modern computers."

Question 21

Question ID: BK05_CH05_EXT_021

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Who was Alan Turing?

Answer Key: A British scientist.

Question 22

Question ID: BK05_CH05_EXT_022

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: During which historical event did Alan Turing build a code-breaking machine?

Answer Key: During World War II.

Question 23

Question ID: BK05_CH05_EXT_023

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What was the primary function of Turing's wartime machine?

Answer Key: To break secret military codes.

Question 24

Question ID: BK05_CH05_EXT_024

Type: Extract Based

Difficulty: Medium

Bloom Level: Understanding

Marks: 1

Question: What foundations did Turing's ideas lay for computing?

Answer Key: Foundations for modern computers and thinking machines (artificial intelligence).

Question 25

Question ID: BK05_CH05_EXT_025

Type: Extract Based

Difficulty: Medium

Bloom Level: Analyzing

Marks: 1

Question: How did Turing's work change the nature of electronic machinery?

Answer Key: He proved that electronic machines could execute complex logical problem-solving.

Extract Set 6

> "The first electronic computer, called ENIAC (Electronic Numerical Integrator and Computer), was built in 1945 by John Presper Eckert and John Mauchly in the United States. It was mammoth, filling an entire room, but it could perform calculations much faster than any human."

Question 26

Question ID: BK05_CH05_EXT_026

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What was the name of the first electronic computer?

Answer Key: ENIAC.

Question 27

Question ID: BK05_CH05_EXT_027

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** In what year was ENIAC built?

Answer Key: 1945.

Question 28

Question ID: BK05_CH05_EXT_028

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who built ENIAC?

Answer Key: John Presper Eckert and John Mauchly in the United States.

Question 29

Question ID: BK05_CH05_EXT_029

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** How large was the ENIAC computer physically?

Answer Key: It was mammoth, filling an entire room.

Question 30

Question ID: BK05_CH05_EXT_030

Type: Extract Based

Difficulty: Medium

Bloom Level: Understanding

Marks: 1

****Question**:** What speed advantage did ENIAC have over human mathematicians?

Answer Key: It performed calculations much faster than any human.

Extract Set 7

> **"Over time, computers became smaller, faster and more affordable. The invention of the microchip in the 1970s made it possible to create personal computers. Companies like Apple, IBM and Microsoft helped bring computers into homes, schools and offices around the world."**

Question 31

Question ID: BK05_CH05_EXT_031

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Which 1970s invention enabled personal computers?

Answer Key: The microchip.

Question 32

Question ID: BK05_CH05_EXT_032

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: How did computers change over time regarding size, speed, and cost?

Answer Key: They became smaller, faster, and more affordable.

Question 33

Question ID: BK05_CH05_EXT_033

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Name three tech companies mentioned that popularized personal computers.

Answer Key: Apple, IBM, and Microsoft.

Question 34

Question ID: BK05_CH05_EXT_034

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Where did personal computers enter people's lives?

Answer Key: In homes, schools, and offices around the world.

Question 35

Question ID: BK05_CH05_EXT_035

Type: Extract Based

Difficulty: Medium

Bloom Level: Analyzing

Marks: 1

Question: Why was the microchip critical for desktop personal computers?

Answer Key: It shrunk millions of electronic circuits onto tiny silicon wafers, reducing hardware size and cost.

Extract Set 8

> "Today, computers are everywhere—on our desks, in our pockets and even in watches! They help us work, learn, play and connect with people worldwide. The invention of the computer truly changed the way we live and work."

Question 36

Question ID: BK05_CH05_EXT_036

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Where can computers be found today according to this passage?

Answer Key: On our desks, in our pockets, and in watches.

Question 37

Question ID: BK05_CH05_EXT_037

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Name four ways computers help humans today.

Answer Key: Help us work, learn, play, and connect with people worldwide.

Question 38

Question ID: BK05_CH05_EXT_038

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** What final statement describes the computer's overall impact?

Answer Key: The invention of the computer truly changed the way we live and work.

Question 39

Question ID: BK05_CH05_EXT_039

Type: Extract Based

Difficulty: Medium

Bloom Level: Applying

Marks: 1

****Question**:** What modern handheld device represents a computer 'in our pockets'?

Answer Key: A smartphone.

Question 40

Question ID: BK05_CH05_EXT_040

Type: Extract Based

Difficulty: Medium

Bloom Level: Evaluating

Marks: 1

****Question**:** Why is the computer considered ubiquitous in modern human society?

Answer Key: Because microprocessor technology is embedded across everyday work, learning, travel, and communication tools.

Extract Set 9

> ****Word Meaning: Integrator: Something which joins things so that they become one. Mammoth: Very big.****

Question 41

Question ID: BK05_CH05_EXT_041

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What is the definition of 'integrator'?

Answer Key: Something which joins things so that they become one.

Question 42

Question ID: BK05_CH05_EXT_042

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: What is the definition of 'mammoth'?

Answer Key: Very big.

Question 43

Question ID: BK05_CH05_EXT_043

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

Question: Which computer name contained the word 'Integrator'?

Answer Key: ENIAC (Electronic Numerical Integrator and Computer).

Question 44

Question ID: BK05_CH05_EXT_044

Type: Extract Based

Difficulty: Medium

Bloom Level: Understanding

Marks: 1

Question: Why was the word 'mammoth' applied to ENIAC?

Answer Key: Because it filled an entire 1,800-square-foot room and weighed 30 tons.

Question 45

Question ID: BK05_CH05_EXT_045

Type: Extract Based

Difficulty: Medium

Bloom Level: Applying

Marks: 1

Question: Use the word 'mammoth' in a complete original sentence.

Answer Key: Constructing the bridge was a mammoth task that took five years.

Extract Set 10

> **"Charles Babbage designed the Difference Engine... Ada Lovelace wrote first algorithms... Alan Turing created a code-breaking machine... ENIAC was built in 1945... Microchips in 1970s made personal computers possible."**

Question 46

Question ID: BK05_CH05_EXT_046

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who designed the Difference Engine?

Answer Key: Charles Babbage.

Question 47

Question ID: BK05_CH05_EXT_047

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who wrote the first computer algorithms?

Answer Key: Ada Lovelace.

Question 48

Question ID: BK05_CH05_EXT_048

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** Who built a code-breaking machine during World War II?

Answer Key: Alan Turing.

Question 49

Question ID: BK05_CH05_EXT_049

Type: Extract Based

Difficulty: Easy

Bloom Level: Remembering

Marks: 1

****Question**:** What hardware breakthrough occurred in the 1970s?

Answer Key: The invention of the microchip.

Question 50

Question ID: BK05_CH05_EXT_050

Type: Extract Based

Difficulty: Hard

Bloom Level: Synthesizing

Marks: 1

****Question**:** What pattern of technological progression is shown across these pioneers?

Answer Key: Progressed from 19th-century mechanical theory to 20th-century electronic engineering and 1970s microchip commercialization.